# AQ5.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Consider a function $T: V \rightarrow W$T : V → W, where $V$V and $W$W are the vector spaces and following statements:

Statement 1:Statement 1:$T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), where $v_1, v_2 \in V$v1, v2 ∈ V.
Statement 2:Statement 2: $T(cv_1) = cT(v_1)$T(cv1) = cT(v1), where $x \in V$x ∈ V and $c \in R$c ∈ R.

If $T$T is a linear transformation then $T$T satisfies
- statement 1 only.
- statement 2 only.
- both the statements 1 and 2.
- neither statement 1 nor statement 2.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
both the statements 1 and 2.
*1 point*
Which of the following is/are a linear transformation.

- [ ] $T: R^2 \rightarrow R^3, T(x, y) = (x, x + y, y + 1)$T : R2 → R3, T(x, y) = (x, x + y, y + 1)
- [ ] $T: R^2 \rightarrow R^2, T(x, y) = (xy, x + y)$T : R2 → R2, T(x, y) = (xy, x + y)
- [ ] $T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y + 1)$T : R2 → R3, T(x, y) = (x + y, y, 2x + y + 1)
- [ ] $T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y)$T : R2 → R3, T(x, y) = (x + y, y, 2x + y)
- [ ] $T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$T : R2 → R2, T(x, y) = (2x, 2y)
- [ ] $T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{xy}{2})$T : R2 → R2, T(x, y) = (2x, 2xy)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T: R^2 \rightarrow R^3, T(x, y) = (x + y, y, 2x + y)$T : R2 → R3, T(x, y) = (x + y, y, 2x + y)
$T: R^2 \rightarrow R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$T : R2 → R2, T(x, y) = (2x, 2y)
*1 point*
Let $T: R^3 \rightarrow R^4$T : R3 → R4 be a linear transformation. Then which of the following options is /are true?

- [ ] $T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), where $v_1, v_2 \in R^3$v1, v2 ∈ R3.
- [ ] $T(0, 0, 0) = (0, 0, 0, 0)$T(0, 0, 0) = (0, 0, 0, 0)
- [ ]

$T(0,0,0) = (1,1,1,1)$T(0,0,0) = (1,1,1,1)

☐

$T(3(1,2,0)) = 3T(1,2,0)$T(3(1,2,0)) = 3T(1,2,0)

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), where $v_1, v_2 \in R^3$v1, v2 ∈ R3.

$T(0,0,0) = (0,0,0,0)$T(0,0,0) = (0,0,0,0)

$T(3(1,2,0)) = 3T(1,2,0)$T(3(1,2,0)) = 3T(1,2,0)

*1 point*

Let $T: R^3 \rightarrow R^2$T : R3 → R2 be a linear transformation, defined as $T(x, y, z) = (x + y, y + z)$ T(x, y, z) = (x + y, y + z). Then which of the following options is /are true?

☐

$T(0,0,0) = (0,0)$T(0,0,0) = (0,0)

☐

$T(1,2,3) = (3,5)$T(1,2,3) = (3,5)

☐

$T(0,1,1) = (1,2)$T(0,1,1) = (1,2)

☐

$T(1,1,1) = (3,2)$T(1,1,1) = (3,2)

☐

$T$T is injective.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T(0,0,0) = (0,0)$T(0,0,0) = (0,0)

$T(1,2,3) = (3,5)$T(1,2,3) = (3,5)

$T(0,1,1) = (1,2)$T(0,1,1) = (1,2)

*1 point*

Choose the correct set of options:

☐

$T$T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$v = 0 ∈ V so that $T(v) = 0$T(v) = 0.

☐

If $T: R^2 \rightarrow R^2$T : R2 → R2 is a surjective linear transformation, then it cannot be injective.

☐

If $T: R \rightarrow R$T : R → R is a linear transformation which is not injective, then $T$T must be 00, i,e., $T(v) = 0$T(v) = 0 for all $v \in R$v ∈ R.

☐

If there exists some non-zero vector $v \in V$v ∈ V , such that $T(v) = 0$T(v) = 0, for a linear transformation $T: V \rightarrow W$T : V → W, then $T$T cannot be an isomorphism.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$v = 0 ∈ V so that $T(v) = 0$T(v) = 0.

If $T: R \rightarrow R$T : R → R is a linear transformation which is not injective, then $T$T must be 00, i,e., $T(v) = 0$T(v) = 0 for all $v \in R$v ∈ R.

If there exists some non-zero vector $v \in V$v ∈ V , such that $T(v) = 0$T(v) = 0, for a linear

transformation $T: V \to W$T : V → W, then $T$T cannot be an isomorphism.

*1 point*

If $T: R^2 \to R^2$T : R2 → R2 is a linear transformation, such that $T(x, y) = (x, 0)$T(x, y) = (x, 0), then which of the following options is true?

○ $T$T is both one to one and onto.

○ $T$T is one to one, but not onto.

○ $T$T is onto, but not one to one.

○ $T$T is neither one to one nor onto.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$T is neither one to one nor onto.

*1 point*

If $T: R^2 \to R^2$T : R2 → R2 is a linear transformation, such that $T(x, y) = (x + y, x - y)$T(x, y) = (x + y, x − y), then which of the following options is true?

○ $T$T is both one to one and onto.

○ $T$T is one to one, but not onto.

○ $T$T is onto, but not one to one.

○ $T$T is neither one to one nor onto.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$T is both one to one and onto.

*1 point*

Let $T: R^2 \to R^2$T : R2 → R2 be a linear transformation, such that $T(2, 1) = (3, 2)$T(2, 1) = (3, 2) and $T(1, 3) = (0, 0)$T(1, 3) = (0, 0). Which of the following options is true?

○ $T(x, y) = \frac{1}{5}(3x, 2y)$T(x, y) = 51(3x, 2y)

○ $T(x, y) = (x + 1, y + 1)$T(x, y) = (x + 1, y + 1)

○ $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$T(x, y) = 51(9x − 3y, 6x − 2y)

○ $T$T cannot be determined from the given information.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$T(x, y) = 51(9x − 3y, 6x − 2y)

*1 point*

Consider the linear transformation $T: R^2 \to R^2$T : R2 → R2 defined as $T(v) = Av$T(v) = Av, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, and $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$v = [xy], and A = [3425]. Which of the following options is true?

○ $T$T is an isomorphism.

○
$T$T is one to one, but not onto.
○
$T$T is onto, but not one to one.
○
$T$T is neither one to one nor onto.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T$T is an isomorphism.

**Level 2**

*1 point*
Let $S: V_1 \rightarrow V_2$S : V1 → V2 and $T: V_2 \rightarrow V_3$T : V2 → V3 be two linear transformations. Let us define $T \circ S: V_1 \rightarrow V_3$T ∘ S : V1 → V3 by $T \circ S(v) = T(S(v))$T ∘ S(v) = T(S(v)). Choose the correct set of options.

☐
If $T \circ S$T ∘ S is injective, then $T$T must be injective.
☐
If $T \circ S$T ∘ S is injective, then $S$S must be injective.
☐
If $T \circ S$T ∘ S is surjective, then $T$T must be surjective.
☐
If $T \circ S$T ∘ S is surjective, then $S$S must be surjective.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If $T \circ S$T ∘ S is injective, then $S$S must be injective.
If $T \circ S$T ∘ S is surjective, then $T$T must be surjective.

*1 point*
If $T: R^2 \rightarrow R^3$T : R2 → R3 is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$T(x, y) = (2x + 3y, 5x − y, x + 6y). Which of the following options is true?

○
$T$T is both one to one and onto.
○
$T$T is one to one, but not onto.
○
$T$T is onto, but not one to one.
○
$T$T is neither one to one nor onto.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T$T is one to one, but not onto.

*1 point*
If $T: R^3 \rightarrow R^3$T : R3 → R3 is a linear transformation defined as follows, $T: R^3 \rightarrow R^3$T : R3 → R3

$T(x, y, z) = (a_1 x + b_1 y + c_1 z, a_2 x + b_2 y + c_2 z, a_3 x + b_3 y + c_3 z)$T(x, y, z) = (a1x + b1y + c1 z, a2x + b2y + c2z, a3x + b3y + c3z). This can be represented as $T(v) = Av$T(v) = Av, where

$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$A = $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$a1a2a3b1b2b3c1c2c3, and $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$v = $\begin{bmatrix} \\ \\ \end{bmatrix}$xyz. Which of the following options are true?

- [ ] $T$T is injective if and only if the system of linear equations $Av = 0$Av = 0 has a unique solution.
- [ ] $T$T is injective if and only if $rank(A) = 3$rank(A) = 3.
- [ ] If the system of linear equations $Av = b$Av = b for any $b \in R^3$b ∈ R3, has a solution, then $T$T must be surjective.
- [ ] If $T$T is not surjective, then there exists some $b \in R^3$b ∈ R3, such that the system of linear equations $Av = b$Av = b has no solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$T is injective if and only if the system of linear equations $Av = 0$Av = 0 has a unique solution.
$T$T is injective if and only if $rank(A) = 3$rank(A) = 3.
If the system of linear equations $Av = b$Av = b for any $b \in R^3$b ∈ R3, has a solution, then $T$T must be surjective.
If $T$T is not surjective, then there exists some $b \in R^3$b ∈ R3, such that the system of linear equations $Av = b$Av = b has no solution.

Check Answers

Your score is: 0/12